

# Shutdown Unit Indicates Incorrect Speed Pickup Fault

## Symptoms

- The SDU410 unit engine speed value reading is correct, but the SDU410 unit has an alarm for malfunctioned engine speed sensor.

## How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

## Shoptalk

The SDU410 unit has two engine speed input signals. If the signals differ, the SDU410 unit uses the higher of the two readings as the signal.

## Troubleshooting Summary

| STEPS   |                                                                                                                           | SPECIFICATIONS     |
|---------|---------------------------------------------------------------------------------------------------------------------------|--------------------|
| STEP 1. | Check customer interface box wiring                                                                                       |                    |
|         | STEP 1A. Check the engine speed 1 signal and return wires for an open.                                                    | Less than 10 ohms? |
|         | STEP 1A-1. Check the engine speed 2 signal and return wires for an open.                                                  | Less than 10 ohms? |
|         | STEP 1B. Check the engine speed 1 signal and return wires for a wire-to-wire short at the SDU410 unit and C4 connector.   | Less than 10 ohms? |
|         | STEP 1B-1. Check the engine speed 2 signal and return wires for a wire-to-wire short at the SDU410 unit and C4 connector. | Less than 10 ohms? |

| STEPS          |                                                                                                                           | SPECIFICATIONS     |
|----------------|---------------------------------------------------------------------------------------------------------------------------|--------------------|
|                | <b>STEP 1C.</b> Check the engine speed 1 signal wire for a short to ground at the SDU410 unit and C4 connector.           | Less than 10 ohms? |
|                | <b>STEP 1C-1.</b> Check the engine speed 2 signal wire for a short to ground at the SDU410 unit and C4 connector.         | Less than 10 ohms? |
| <b>STEP 2.</b> | Check the OEM wiring harness                                                                                              |                    |
|                | <b>STEP 2A.</b> Check the engine speed 1 signal and return wires for an open at the C4 and C11 connectors.                | Less than 10 ohms? |
|                | <b>STEP 2A-1.</b> Check the engine speed 2 signal and return wires for an open at the C4 and C11 connectors.              | Less than 10 ohms? |
|                | <b>STEP 2B.</b> Check the engine speed 1 signal and return wires for a wire-to-wire short at the C4 and C11 connectors.   | Less than 10 ohms? |
|                | <b>STEP 2B-1.</b> Check the engine speed 2 signal and return wires for a wire-to-wire short at the C4 and C11 connectors. | Less than 10 ohms? |
|                | <b>STEP 2C.</b> Check the engine speed 1 signal wire for a short to ground at the C4 and C11 connectors.                  | Less than 10 ohms? |
|                | <b>STEP 2C-1.</b> Check the engine speed 2 signal wire for a short to ground at the C4 and C11 connectors.                | Less than 10 ohms? |

**STEP 1. Check the customer interface box wiring.**

## STEP 1A. Check the engine speed 1 signal and return wires for an open.

### Conditions :

- Open the customer interface box.
- Disconnect the engine speed 1 signal and return wires from the SDU410 unit and the C4 connector.

| Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Specification/Repair                                                                                                                                                                                                    | Next Step                     |
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| <p>Check the signal and return wires for an open.</p> <ul style="list-style-type: none"><li>• Place one test lead on the engine speed 1 supply wire at the SDU410 unit.</li><li>• Place the other test lead on the engine speed 1 signal pin at the C4 connector.</li><li>• Place one test lead on the engine speed 1 return wire at the SDU410 unit.</li><li>• Place the other test lead on the engine speed 1 return pin at the C4 connector.</li></ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>1A-1</b></p>            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

## STEP 1A-1. Check the engine speed 2 signal and return wires for an open.

### Conditions :

- Open the customer interface box.
- Disconnect the engine speed 2 signal and return wires from the SDU410 unit and the C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 2 supply wire at the SDU410 unit.</li> <li>Place the other test lead on the engine speed 2 signal pin at the C4 connector.</li> <li>Place one test lead on the engine speed 2 return wire at the SDU410 unit.</li> <li>Place the other test lead on the engine speed 2 return pin at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <b>1B</b>              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <b>Repair complete</b> |

**STEP 1B. Check the engine speed 1 signal and return wires for a wire-to-wire short at the SDU410 unit and C4 connector.**

**Conditions :**

- Open the customer interface box.
- Disconnect the engine speed 1 signal and return wires from the SDU410 unit and disconnect the C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 1 signal wire at the SDU410 unit.</li> <li>Place the other test lead on all other wires on the SDU410 unit.</li> <li>Place one test lead on the engine speed 1 return wire at the SDU410 unit.</li> <li>Place the other test lead on all other wires on the SDU410 unit.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1B-1</b></p>            |

**STEP 1B-1. Check the engine speed 2 signal and return wires for a wire-to-wire short at the SDU410 unit and C4 connector.**

**Conditions :**

- Open the customer interface box.
- Disconnect the engine speed 2 signal and return wires from the SDU410 unit and disconnect the C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 2 supply wire at the SDU410 unit.</li> <li>Place the other test lead on all other wires at the SDU410 unit.</li> <li>Place one test lead on the engine speed 2 return wire at the SDU410 unit.</li> <li>Place the other test lead on all other wires at the SDU410 unit.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1C</b></p>              |

**STEP 1C. Check the engine speed 1 signal wire for a short to ground at the SDU410 unit and C4 connector.**

**Conditions :**

- Open the customer interface box.
- Disconnect the engine speed 1 signal wire from the SDU410 unit and disconnect the C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the signal wire for short to ground.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 1 signal at the SDU410 unit.</li> <li>Place the other test lead to panel ground.</li> <li>Place one test lead on the engine speed 1 signal pin at the C4 connector.</li> <li>Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>1C-1</b></p>            |

**STEP 1C-1. Check the engine speed 2 signal wire for a short to ground at the SDU410 unit and C4 connector.**

**Conditions :**

- Open the customer interface box.
- Disconnect the engine speed 2 signal wire from the SDU410 unit and disconnect the C4 connector.

| Action | Specification/Repair | Next Step |
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| <p>Check the signal wire for short to ground.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 2 signal at the SDU410 unit.</li> <li>Place the other test lead to panel ground.</li> <li>Place one test lead on the engine speed 1 signal pin at the C4 connector.</li> <li>Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>2A</b></p>              |

## STEP 2. Check the OEM wiring harness.

### STEP 2A. Check the engine speed 1 signal and return wires for an open at the C4 and C11 connectors.

**Conditions :**

- Open the customer interface box.
- Disconnect the C4, C11, and engine speed 1 sensor connectors.

| Action | Specification/Repair | Next Step |
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| <p>Check the signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>• Place one test lead on the engine speed 1 signal pin at the C4 connector.</li> <li>• Place the other test lead on the engine speed 1 signal pin at the C11 connector.</li> <li>• Place one test lead on the engine speed 1 return pin at the C4 connector.</li> <li>• Place the other test lead on the engine speed 1 return pin at the C11 connector.</li> <li>• Place one test lead on the engine speed 1 signal pin at the sensor connector.</li> <li>• Place the other test lead on the engine speed 1 signal pin at the C11 connector.</li> <li>• Place one test lead on the engine speed 1 return pin at the sensor connector.</li> <li>• Place the other test lead on the engine speed 1 return pin at the C11 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <hr/> <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>2A-1</b></p> <hr/> <p><b>Repair complete</b></p> |
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**STEP 2A-1. Check the engine speed 2 signal and return wires for an open at the C4 and C11 connectors.**

**Conditions :**

- Open the customer interface box.
- Disconnect the C4, C11, and engine speed 2 sensor connectors.

**Action****Specification/Repair****Next Step**

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| <p>Check the signal and return wires for an open.</p> <ul style="list-style-type: none"> <li>• Place one test lead on the engine speed 2 signal pin at C4 connector.</li> <li>• Place the other test lead on the engine speed 2 signal pin at the C11 connector.</li> <li>• Place one test lead on the engine speed 2 return pin at the C4 connector.</li> <li>• Place the other test lead on the engine speed 2 return pin at the C11 connector.</li> <li>• Place one test lead on the engine speed 2 signal pin at the sensor connector.</li> <li>• Place the other test lead on the engine speed 2 signal pin at the C11 connector.</li> <li>• Place one test lead on the engine speed 2 return pin at the sensor connector.</li> <li>• Place the other test lead on the engine speed 2 return pin at the C11 connector.</li> </ul> <p>Refer to the circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                                                             | <p><b>2B</b></p>              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |

**STEP 2B. Check the engine speed 1 signal and return wires for a wire-to-wire short at the C4 and C11 connectors.**

**Conditions :**

- Open the customer interface box.
- Disconnect the C4 and C11 connectors.

| Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Specification/Repair                                                                                                                                                                                                     | Next Step                     |
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| <p>Check the signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"><li>• Place one test lead on the engine speed 1 signal pin at the C4 connector.</li><li>• Place the other test lead on all other pins at the C4 connector.</li><li>• Place one test lead on the engine speed 1 return pin at the C4 connector.</li><li>• Place the other test lead on all other pins at the C4 connector.</li></ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>2B-1</b></p>            |

**STEP 2B-1. Check the engine speed 2 wire signal and return for a wire-to-wire short at the C4 and C11 connectors.****Conditions :**

- Open the customer interface box.
- Disconnect the C4 and C11 connectors.

| Action | Specification/Repair | Next Step |
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| <p>Check the signal and return wires for a wire-to-wire short.</p> <ul style="list-style-type: none"> <li>Place one test lead on the engine speed 2 signal pin at the C4 connector.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> <li>Place one test lead on the engine speed 2 return pin at the C4 connector.</li> <li>Place the other test lead on all other pins at the C4 connector.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                               | <p><b>2C</b></p>              |

**STEP 2C. Check the engine speed 1 signal wire for a short to ground at the C4 and C11 connectors.**

**Conditions :**

- Open the customer interface box.
- Disconnect the C4, C11, and engine speed 1 sensor connectors.

| Action | Specification/Repair | Next Step |
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| <p>Check the signal wire for short to ground.</p> <ul style="list-style-type: none"> <li>• Place one test lead on the engine speed 1 signal pin at the C4 connector.</li> <li>• Place the other test lead to panel ground.</li> <li>• Place one test lead on the engine speed 1 signal pin at the C11 connector.</li> <li>• Place the other test lead on panel ground.</li> <li>• Place one test lead on the engine speed 1 signal pin at the engine speed 1 sensor connector.</li> <li>• Place the other test lead on panel ground.</li> </ul> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
| <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Less than 10 ohms?</p> <p><b>NO</b></p>                                                                                                                                                                           | <p><b>2C-1</b></p>            |

**STEP 2C-1. Check the engine speed 2 signal wire for a short to ground at the C4 and C11 connectors.**

| <p><b>Conditions :</b></p> <ul style="list-style-type: none"> <li>• Open the customer interface box.</li> <li>• Disconnect the C4, C11, and engine speed 2 sensor connectors.</li> </ul> |                      |           |
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| Action                                                                                                                                                                                   | Specification/Repair | Next Step |

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| <p>Check the signal wire for short to ground.</p> <ul style="list-style-type: none"> <li>• Place one test lead on the engine speed 2 signal pin at the C4 connector.</li> <li>• Place the other test lead to panel ground.</li> <li>• Place one test lead on the engine speed 2 signal pin at the C11 connector.</li> <li>• Place the other test lead on panel ground.</li> <li>• Place one test lead on the engine speed 2 signal pin at the engine speed 2 sensor connector.</li> <li>• Place the other test lead on panel ground.</li> </ul> <p>Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.</p> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.<br/>(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)</p> | <p><b>Repair complete</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>The troubleshooting steps <b>must</b> be checked again from the beginning. A fault mode should have been detected.</p>                              | <p><b>1A</b></p>              |